Data Point

i	x	y
X1	2	6
X2	3	4
X3	3	8
X4	4	7
X5	6	2
X6	6	4
X7	7	3
X8	7	4
X9	8	5
X10	7	6

i	x	y
X1	2	6
X2	3	4
X3	3	8
X4	4	7
X5	6	2
X6	6	4
X7	7	3
X8	7	4
X9	8	5
X10	7	6

- Apply K-Medoid clustering algorithm to form two clusters.
- Use Manhattan distance to find the between data point and medoid.

Step 1

- Select two medoids
- C1=(3, 4)
- C2=(7, 4)

i	x	y	C1	C2	Cluster
X1	2	6			
X2	3	4			
X3	3	8			
X4	4	7			
X5	6	2			
X6	6	4			
X7	7	3			
X8	7	4			
X9	8	5			
X10	7	6			

Step 1

- Select two medoids
- C1=(3, 4)
- C2=(7, 4)
- (x_1, y_1) *and* (x_2, y_2) *are data points*
- $\boldsymbol{Manhattan\ Dist} = |\boldsymbol{x_1} - \boldsymbol{x_2}| + |\boldsymbol{y_1} - \boldsymbol{y_2}|$

i	x	y	C1	C2	Cluster
X1	2	6			
X2	3	4			
X3	3	8			
X4	4	7			
X5	6	2			
X6	6	4			
X7	7	3			
X8	7	4			
X9	8	5			
X10	7	6			

Step 1

- Select two medoids
- C1=(3, 4)
- C2=(7, 4)
- $\boldsymbol{Manhattan\ Dist} = |\boldsymbol{x_1} - \boldsymbol{x_2}| + |\boldsymbol{y_1} - \boldsymbol{y_2}|$
- $\boldsymbol{Mdist}[(\mathbf{2}, \mathbf{6}), (\mathbf{3}, \mathbf{4})] = |\mathbf{2} - \mathbf{3}| + |\mathbf{6} - \mathbf{4}| = \mathbf{3}$
- $\boldsymbol{Mdist}[(\mathbf{3}, \mathbf{4}), (\mathbf{3}, \mathbf{4})] = |\mathbf{3} - \mathbf{3}| + |\mathbf{4} - \mathbf{4}| = \mathbf{0}$

i	x	y	C1	C2	Cluster
X1	2	6	3		
X2	3	4	0		
X3	3	8			
X4	4	7			
X5	6	2			
X6	6	4			
X7	7	3			
X8	7	4			
X9	8	5			
X10	7	6			

Step 1

- Select two medoids
- C1=(3, 4)
- C2=(7, 4)
- $\boldsymbol{Manhattan\ Dist} = |\boldsymbol{x_1} - \boldsymbol{x_2}| + |\boldsymbol{y_1} - \boldsymbol{y_2}|$
- $\boldsymbol{Mdist}[(\mathbf{2},\mathbf{6}),(\mathbf{3},\mathbf{4})] = |\mathbf{2}-\mathbf{3}| + |\mathbf{6}-\mathbf{4}| = \mathbf{3}$
- $\boldsymbol{Mdist}[(\mathbf{3},\mathbf{4}),(\mathbf{3},\mathbf{4})] = |\mathbf{3}-\mathbf{3}| + |\mathbf{4}-\mathbf{4}| = \mathbf{0}$

i	x	y	C1	C2	Cluster
X1	2	6	3	7	C1
X2	3	4	0	4	C1
X3	3	8	4	8	C1
X4	4	7	4	6	C1
X5	6	2	5	3	C2
X6	6	4	3	1	C2
X7	7	3	5	1	C2
X8	7	4	4	0	C2
X9	8	5	6	2	C2
X10	7	6	6	2	C2

- C1: {(2,6), **(3,4),** (3,8), (4,7)}
- C2: {(6, 2), (6, 4), (7, 3), **(7, 4),** (8, 5), (7,6)}
- **Calculate the Total Cost**
- $\boldsymbol{Cost(c,x)} = \sum_i |\boldsymbol{c_i} - \boldsymbol{x_i}|$

- C1: {(2,6), **(3,4),** (3,8), (4,7)}
- C2: {(6, 2), (6, 4), (7, 3), **(7, 4),** (8, 5), (7,6)}
- **Calculate the Total Cost**
- $\boldsymbol{Cost}(\boldsymbol{c}, \boldsymbol{x}) = \sum_i |\boldsymbol{c}_i - \boldsymbol{x}_i|$
- $Total\ Cost = \{Cost\big((3,4),(2,6)\big) + Cost\big((3,4),(3,8)\big) + Cost\big((3,4),(4,7)\big) + Cost\big((7,4),(6,2)\big) + Cost\big((7,4),(6,4)\big) + Cost\big((7,4),(7,3)\big) + Cost\big((7,4),(8,5)\big) + Cost\big((7,4),(7,6)\big)\}$

- C1: {(2,6), **(3,4),** (3,8), (4,7)}
- C2: {(6, 2), (6, 4), (7, 3), **(7, 4),** (8, 5), (7,6)}
- **Calculate the Total Cost**
- $\boldsymbol{Cost(c, x)} = \sum_i |\boldsymbol{c_i} - \boldsymbol{x_i}|$
- $Total\ Cost = \{Cost\big((3,4),(2,6)\big) + Cost\big((3,4),(3,8)\big) + Cost\big((3,4),(4,7)\big) + Cost\big((7,4),(6,2)\big) + Cost\big((7,4),(6,4)\big) + Cost\big((7,4),(7,3)\big) + Cost\big((7,4),(8,5)\big) + Cost\big((7,4),(7,6)\big)\}$
- $\boldsymbol{Total\ Cost} = \mathbf{3 + 4 + 4 + 2 + 3 + 1 + 1 + 2 = 20}$

Step 3

- **Randomly select one non-medoid point and recalculate the cost.**
- C1=(3, 4) and C2=(7, 4)
- **O=(7, 3)**
- **Swap C2 with O**
- **New Medoids**
- **C1=(3, 4) and O=(7, 3)**

i	x	y	C1	O	Clust
X1	2	6			
X2	3	4			
X3	3	8			
X4	4	7			
X5	6	2			
X6	6	4			
X7	7	3			
X8	7	4			
X9	8	5			
X10	7	6			

Step 3

- **New Medoids**
- C1=(3, 4) and O=(7, 3)
- $Manhattan\ Dist = |x_1 - x_2| + |y_1 - y_2|$

- $Mdist[(2,6),(7,3)] = |2-7| + |6-3| = 8$

i	x	y	C1	O	Cluster
X1	2	6	3		
X2	3	4	0		
X3	3	8	4		
X4	4	7	4		
X5	6	2	5		
X6	6	4	3		
X7	7	3	5		
X8	7	4	4		
X9	8	5	6		
X10	7	6	6		

Step 3

- **New Medoids**
- C1=(3, 4) and O=(7, 3)
- $\boldsymbol{Manhattan\ Dist} = |x_1 - x_2| + |y_1 - y_2|$
- $\boldsymbol{Mdist}[(2,6),(7,3)] = |2-7| + |6-3| = 8$

i	x	y	C1	O	Cluster
X1	2	6	3	8	
X2	3	4	0	5	
X3	3	8	4	9	
X4	4	7	4	7	
X5	6	2	5	2	
X6	6	4	3	2	
X7	7	3	5	0	
X8	7	4	4	1	
X9	8	5	6	3	
X10	7	6	6	3	

Step 3

- **New Medoids**
- **C1=(3, 4) and O=(7, 3)**
- $Manhattan\ Dist\ = |x_1 - x_2| + |y_1 - y_2|$
- $Mdist[(2,6),(7,3)] = |2-7| + |6-3|$ =8

i	x	y	C1	O	Cluster
X1	2	6	3	8	C1
X2	3	4	0	5	C1
X3	3	8	4	9	C1
X4	4	7	4	7	C1
X5	6	2	5	2	O
X6	6	4	3	2	O
X7	7	3	5	0	O
X8	7	4	4	1	O
X9	8	5	6	3	O
X10	7	6	6	3	O

- C1: {(2,6), **(3,4),** (3,8), (4,7)}
- O: {(6, 2), (6, 4), **(7, 3),** (7, 4), (8, 5), (7,6)}
- **Calculate the Total Cost**
- $\boldsymbol{Cost(c,x)} = \sum_i |\boldsymbol{c_i} - \boldsymbol{x_i}|$
- $Current\ Total\ Cost = \{Cost\big((3,4),(2,6)\big) + Cost\big((3,4),(3,8)\big) + Cost\big((3,4),(4,7)\big) + Cost\big((7,3),(6,2)\big) + Cost\big((7,3),(6,4)\big) + Cost\big((7,3),(7,4)\big) + Cost\big((7,3),(8,5)\big) + Cost\big((7,3),(7,6)\big)\}$
- $\boldsymbol{Current\ Total\ Cost} = \mathbf{3 + 4 + 4 + 2 + 2 + 1 + 3 + 3} \cancel{=} \mathbf{22}$

Step 4

- Cost of Swapping of medoid C2 with O
- $S = Current\ Total\ Cost - Previous\ Total\ Cost$
- $S = 22 - 20 = 2 > 0$
- Hence Swapping C2 with O is not a good Idea.
- Final Medoids are C1 =(3, 4) and C2=(7, 4)